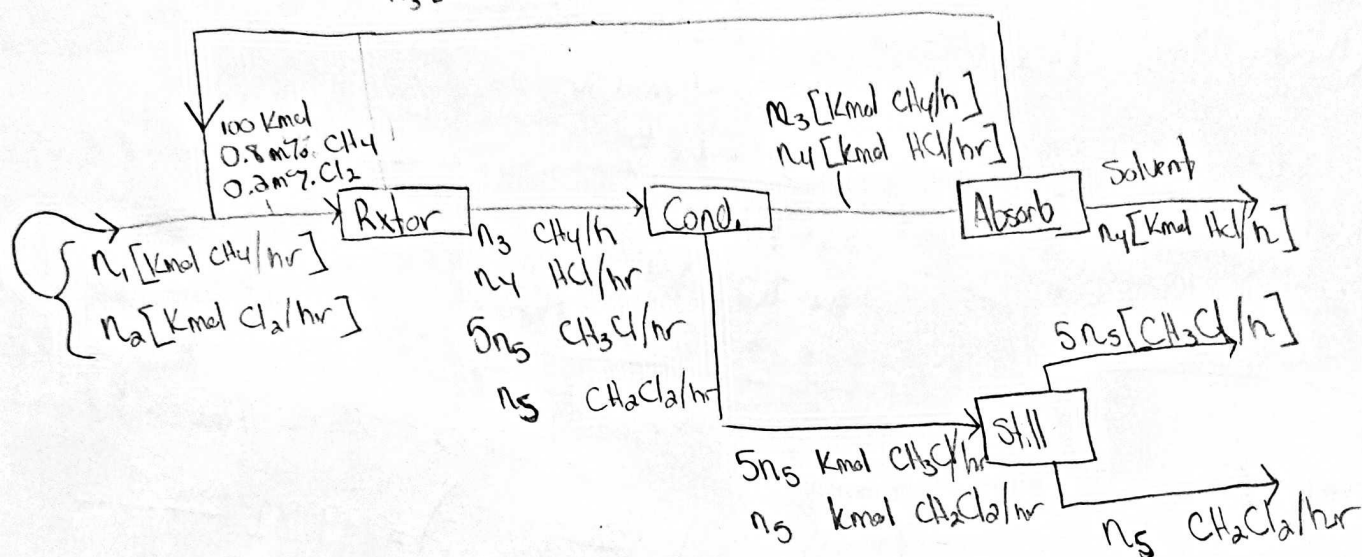


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1) Chem 210 Homework #8 Mark Hurley

n_3 [Kmol CH_4/h]



A. Basis 100 kmol/h entering reactor.

DoF Overall

Unks = 4 { n_1, n_2, n_5, n_4 }
Balances on C, H, Cl possible
 $4 - 3 = 1$

DoF Mixing Entry Pt.

Unks = n_1, n_2, n_3 3-2 = 1
Balances on CH_4, Cl_2

DoF Condensor

Unks = n_3, n_4, n_5 3-0 = 3
no equations

DoF Reactor

Unks = n_3, n_4, n_5
Balances on C, H, Cl
 $3 - 3 = 0$

DoF Absorber

Unks. = n_3, n_4
no equations
 $2 - 0 = 2$

DoF Distillation Column

Unks. = n_4, n_5 2-0 = 2
0 equations

Rxtor

- (1) $n_1 + n_3 = 80$
(2) $n_2 = 20$
(3) H balance. $4n_3 + 3 \cdot 5n_5 + n_4 + 2 \cdot n_5 = 4 \cdot 80$
(4) C balance $n_3 + n_5 + 5n_5 = 80$
(5) Cl Balance $n_4 + 5n_5 + 2n_5 = 20 \cdot 2$

inflow

1.)

$$4(n_3 + 5n_5 + n_6 = 80) \text{ (eq 4)} =$$

$$-1(4n_3 + n_4 + 17n_5 = 320) \text{ (eq 3)} =$$

$$(n_4 + 5n_5 + 2n_6 = 40) \text{ (eq 5)} =$$

$$4n_3 + 24n_5 = 320 \text{ kmol/hr}$$

$$-4n_3 - 17n_5 - n_4 = -320 \text{ kmol/hr}$$

$$7n_5 + n_4 = 40 \text{ kmol/hr}$$

$$14n_5 = 40 \text{ kmol/hr}$$

$$n_5 = 2.86 \text{ kmol/hr} \quad \therefore 5n_5 = 14.3 \text{ kmol/hr.}$$

$$n_4 + 7n_5 = 40$$

$$n_4 + 7(2.86) = 40$$

$$n_4 = \sim 20 \text{ kmol/hr.}$$

$$n_3 + 6n_5 = 80$$

$$n_3 + 6(2.86) = 80$$

$$n_3 = 62.84 \text{ kmol/hr.}$$

$$n_1 + n_3 = 80$$

$$n_1 = 80 - 62.84$$

$$n_1 = 17.16 \text{ kmol/hr}$$

C. CH_3Cl $12 + 3 + 35.43 = 50.43 \text{ g/mol} = \text{kg/kmol}$

$$\left(\frac{1000 \text{ kg CH}_3\text{Cl}}{\text{hr}} \right) \left(\frac{1 \text{ kmol}}{50.43 \text{ kg/kmol}} \right) = 19.81 \frac{\text{kmol CH}_3\text{Cl}}{\text{hr}}$$

$$\text{Calc Scale Factor} = \frac{19.81}{(5n_5) 14.3} = 1.385$$

$$n_{4SF} = 17.16 \cdot SF(1.385) = 23.766 \text{ kmol CH}_4/\text{hr}$$

$$n_{2SF} = 20 \cdot SF(1.385) = 27.7 \text{ kmol Cl}_2/\text{hr}$$

$$n_{3SF} = 62.84 \cdot SF(1.385) = 87.03 \text{ CH}_4/\text{hr}$$

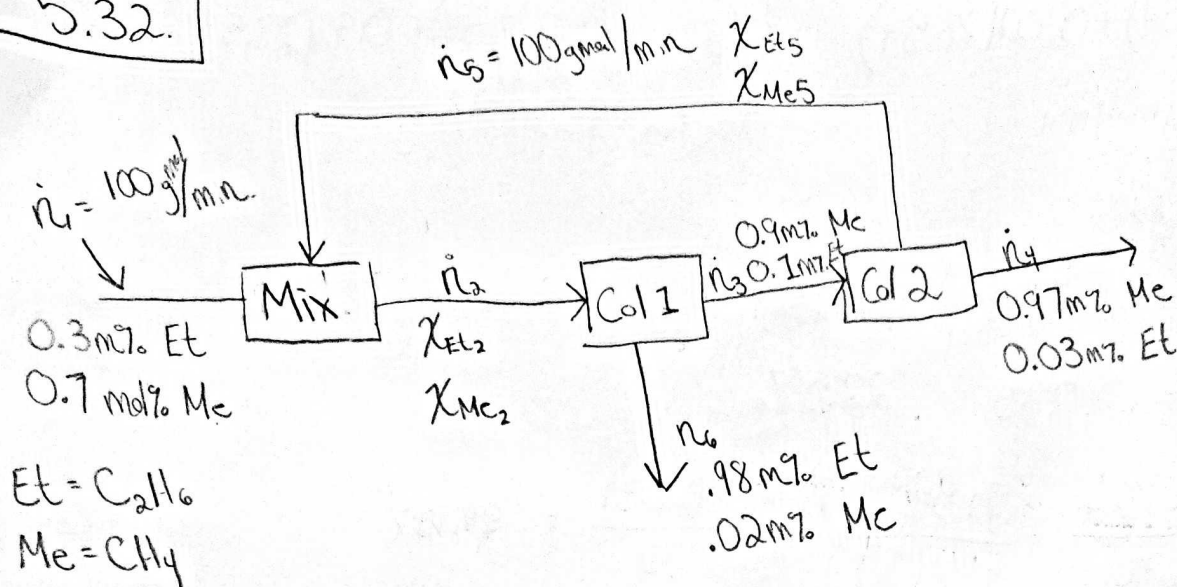
$$\text{Fresh Feed} = n_1 + n_2 = 23.76 + 27.7$$

$$= 51.46 \text{ kmol/hr}$$

$$X_{1SF} = \frac{23.76}{51.46} = 46.17\% \text{ CH}_4$$

$$X_{2SF} = \frac{27.7}{51.46} = 53.87\% \text{ Cl}_2$$

5.32.



Eqs

Mixer

$$\begin{aligned} \text{Me } 70 \text{ g/mol/min} + (100)(x_{Me5}) &= n_2(x_{Me2}) \\ \text{Et } 30 + (100)(x_{Et5}) &= n_2(x_{Et2}) \end{aligned}$$

Col 2

$$\begin{aligned} \text{Me } 0.9n_3 &= (100)(x_{Me5}) + 0.97n_4 \\ \text{Et } 0.1n_3 &= (100)(x_{Et5}) + 0.03n_4 \end{aligned}$$

Stinks...
6 eqs...

$$70 = n_2 \cdot x_{Me2} - (100)x_{Me5}$$

$$70 = (0.9n_3 + 0.02n_6) - (0.9n_3 - 0.97n_4)$$

$$\begin{aligned} * 70 &= 0.97n_4 + 0.02n_6 \\ 70 &= 0.97n_4 + 0.02n_6 \\ - \frac{1}{49}(30 &= 0.98n_6 - 0.03n_4) \\ - \frac{30}{49} &= 0.02n_6 + \frac{0.03}{49}n_4 \end{aligned}$$

$$\begin{aligned} 69.39 &= 0.97n_4 \\ n_4 &= 71.54 \text{ g/mol/min} \\ \therefore n_6 &= 28.59 \text{ g/mol/min} \end{aligned}$$

Col 1

$$\begin{aligned} \text{Me } n_2 \cdot x_{Me2} &= 0.9n_3 + 0.02n_6 \\ \text{Et } n_2 \cdot x_{Et2} &= 0.1n_3 + 0.98n_6 \end{aligned}$$

Need 2 more equations

$$x_{Me5} + x_{Et5} = 1$$

$$x_{Me2} + x_{Et2} = 1$$

$$30 = n_2 \cdot x_{Et2} - x_{Et5}(100)$$

$$30 = (0.1n_3 + 0.98n_6) - (0.1n_3 - 0.03n_4)$$

$$* 30 = 0.98n_6 - 0.03n_4$$

$$n_3 = n_4 + n_5$$

$$\begin{aligned} n_3 &= 71.54 + 100 \\ n_3 &= 171.54 \text{ g/mol/min} \end{aligned}$$

$$0.9(171.54) = 100x_{Me5} + 0.97(71.54)$$

$$x_{Me5} = 85\%$$

$$\therefore x_{Et5} = 15\%$$

5.32

$$x_{Me_2} \dot{n}_2 = 0.9(171.54) + 0.02(28.59)$$

$$x_{Me_2} \dot{n}_2 = 155 \text{ g mol/min}$$

$$\dot{n}_2 = 155 + 45 = 200 \text{ g mol/min}$$

$$x_{Et_2} \dot{n}_2 = 0.1(171.54) + 0.98(28.59)$$

$$x_{Et_2} \dot{n}_2 = 45 \text{ g mol/min}$$

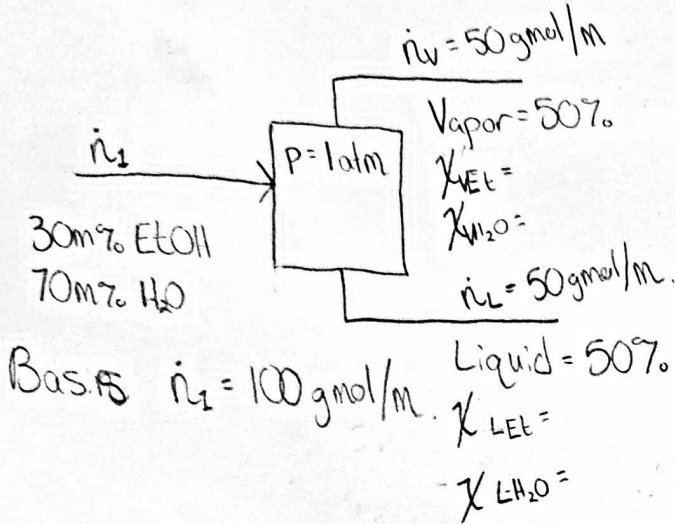
$$x_{Me_2} = 155/200 = 0.775 \text{ or } 77.5\%$$

$$x_{Et_2} = 45/200 = 0.225 \quad 22.5\%$$

$$f_{RMe_4} = \frac{x_{Me_4} \dot{n}_4}{x_{Me_1} \dot{n}_1} = \frac{71.54(0.97)}{100(0.7)} = \frac{69.39}{70} = 99.13\%$$

$$f_{REt_6} = \frac{x_{Et_6} \dot{n}_6}{x_{Et_1} \dot{n}_1} = \frac{28.4(0.98)}{100(0.3)} = \frac{27.832}{30} = 92.77\%$$

5.34
B-11 pg 657



Eqs

$$\dot{n}_L \chi_{LEt} (50) = \dot{n}_V \chi_{VLEt} (50) + \dot{n}_L \chi_{LEt} (50)$$

$$\dot{n}_L \chi_{LH_2O} (50) = \dot{n}_V \chi_{VH_2O} (50) + \dot{n}_L \chi_{LEt} (50)$$

$$\chi_{VLEt} + \chi_{VH_2O} = 1$$

$$\chi_{LEt} + \chi_{LH_2O} = 1$$

Using Excel to plug values from Table B-11

Temp °C	χ_{LEt}	χ_{VLEt}
82.7	0.2337	0.5445
84.1	0.1661	0.5089
85.3	0.01238	0.4704

$50(0.4704) + 50(0.1238) = 29.71 \rightarrow \text{goal was } 30.$
 Temp just above 85.3 °C

$$f_{REtV} = \frac{(50)(0.4704)}{(100)(0.3)} = 0.78$$

78.37%

$$f_{RH_2OL} = \frac{(0.876)(50)}{(0.7)(100)} = 0.626$$

62.6%

$\chi_w = 1 - \chi_E$
 $1 - 0.1238$
 $\chi_w = 0.876$

Sep factor = $\frac{(0.47)(0.876)}{(0.124)(0.53)} = 6.26$

$(\chi_{VLEt} \chi_{Lw}) \uparrow 1 - 0.407 = 0.529$
 $(\chi_{LEt} \chi_{LH_2O})$

Comment Summary